

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, rates, proportional relationships, and units

04

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

A wind turbine completes 900 revolutions in 50 minutes. At this rate, how many revolutions per minute does this turbine complete?

- A. 18
- B. 850
- C. 950
- D. 1,400

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

An object's speed is 64 yards per second. What is the object's speed, in feet per second? 1 yard = 3 feet

- A. 61
- B. 67
- C. 94
- D. 192

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A giant armadillo has a mass of 39 kilograms. What is the giant armadillo's mass in **grams**? (1 kilogram = 1,000 grams)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many feet are equivalent to 34 yards? 1 yard = 3 feet

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 1,116 inches? 1 yard = 36 inches

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

The ratio of the length of line segment XY to the length of line segment ZV is 6 to 1. If the length of line segment XY is 102 inches, what is the length, in inches, of line segment ZV ?

- A. 17
- B. 96
- C. 102
- D. 612

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

The number of raccoons in a 131-square-mile area is estimated to be 2,358. What is the estimated population density, in raccoons per square mile, of this area?

- A. 18
- B. 131
- C. 149
- D. 2,376

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

Shaquan has 7 red cards and 28 blue cards. What is the ratio of red cards to blue cards that Shaquan has?

- A. 1 to 4
- B. 4 to 1
- C. 1 to 7
- D. 7 to 1

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

A cherry pitting machine pits 12 pounds of cherries in 3 minutes. At this rate, how many minutes does it take the machine to pit 96 pounds of cherries?

- A. 8
- B. 15
- C. 24
- D. 36

Q10

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A mechanical device in a workshop produces items at a constant rate of 60 items per hour. At this rate, how many items will the mechanical device produce in 3 hours?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.